

Mathematics-2
Tutorial Sheet-4

Question 1

Given that the vector space P_2 consisting of all polynomials of degree at most two together with the zero polynomial.

We have

$$S = \{-t^2 + 3, -2t^2 - 2t + 3\}$$

a set of polynomials in P_2 .

We have to show that the set S is linearly independent in P_2 .

We know that, the Wronskian of two functions $f_1(t)$ and $f_2(t)$ is given by:

$$W(f_1, f_2)(t) = \begin{vmatrix} f_1(t) & f_2(t) \\ f_1'(t) & f_2'(t) \end{vmatrix}$$

The given polynomials are:

$$f_1(t) = -t^2 + 3, \quad f_2(t) = -2t^2 - 2t + 3$$

$$f_1'(t) = \frac{d}{dt}(-t^2 + 3) = -2t$$

$$f_2'(t) = \frac{d}{dt}(-2t^2 - 2t + 3) = -4t - 2$$

$$W(f_1, f_2)(t) = \begin{vmatrix} -t^2 + 3 & -2t^2 - 2t + 3 \\ -2t & -4t - 2 \end{vmatrix}$$

$$\begin{aligned} &= (-t^2 + 3)(-4t - 2) - (-2t^2 - 2t + 3)(-2t) \\ &= (4t^3 + 2t^2 - 12t - 6) - (4t^3 + 4t^2 - 6t) \\ &= 4t^3 + 2t^2 - 12t - 6 - 4t^3 - 4t^2 + 6t \\ &= -2t^2 - 6t - 6 \end{aligned}$$

Since

$$W(f_1, f_2)(t) = -2t^2 - 6t - 6$$

is a nonzero polynomial, the given set S is linearly independent in P_2 .

Question 2

(a) $\{e^x, e^x \sin x\}$

Consider $f_1(x) = e^x$ and $f_2(x) = e^x \sin x$.

The Wronskian is given by:

$$W(f_1, f_2)(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix}.$$

Computing derivatives:

$$f_1'(x) = e^x, \quad f_2'(x) = e^x \cos x + e^x \sin x.$$

Now, the Wronskian is

$$W(f_1, f_2)(x) = \begin{vmatrix} e^x & e^x \sin x \\ e^x & e^x (\cos x + \sin x) \end{vmatrix}.$$

$$\implies W(f_1, f_2)(x) = e^x \cdot [e^x (\cos x + \sin x)] - e^x \cdot (e^x \sin x).$$

$$\implies W(f_1, f_2)(x) = e^{2x} (\cos x + \sin x - \sin x) = e^{2x} \cos x.$$

Since $W(f_1, f_2)(x) \neq 0$ for all x , the functions are linearly independent.

(b) $\{x \ln |x|^2, x^2 \ln |x|^2\}$

Given the functions:

$$f_1(x) = x \ln |x|^2, \quad f_2(x) = x^2 \ln |x|^2$$

We know

$$\ln |x|^2 = 2 \ln |x|$$

So the functions

$$f_1(x) = 2x \ln |x|, \quad f_2(x) = 2x^2 \ln |x|$$

$$f_1'(x) = \frac{d}{dx}(2x \ln |x|) = 2(\ln |x| + 1)$$

$$f_2'(x) = \frac{d}{dx}(2x^2 \ln |x|) = 2(2x \ln |x| + x) = 4x \ln |x| + 2x$$

The Wronskian is:

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix} = \begin{vmatrix} 2x \ln |x| & 2x^2 \ln |x| \\ 2(\ln |x| + 1) & 4x \ln |x| + 2x \end{vmatrix}$$

$$\begin{aligned}
\implies W(x) &= (2x \ln |x|)(4x \ln |x| + 2x) - (2x^2 \ln |x|)(2(\ln |x| + 1)) \\
&= 8x^2 \ln^2 |x| + 4x^2 \ln |x| - 4x^2 \ln^2 |x| - 4x^2 \ln |x| \\
&= 4x^2 \ln^2 |x|
\end{aligned}$$

Since

$$W(x) = 4x^2 \ln^2 |x|$$

is not identically zero (it is non-zero for $x \neq 0$ and $x \neq \pm 1$), the functions $f_1(x)$ and $f_2(x)$ are linearly independent.

(c) $\{1, e^x\}$

$$W = \begin{vmatrix} 1 & e^x \\ 0 & e^x \end{vmatrix} = e^x \neq 0$$

Thus, the functions are linearly independent.

(d) $\{e^x, \sin x, \cos x\}$

Given functions:

$$f_1 = e^x, \quad f_2 = \sin x, \quad f_3 = \cos x$$

$$f'_1 = e^x, \quad f'_2 = \cos x, \quad f'_3 = -\sin x$$

$$f''_1 = e^x, \quad f''_2 = -\sin x, \quad f''_3 = -\cos x$$

$$W = \begin{vmatrix} e^x & \sin x & \cos x \\ e^x & \cos x & -\sin x \\ e^x & -\sin x & -\cos x \end{vmatrix}$$

$$\implies W = e^x \begin{vmatrix} \cos x & -\sin x \\ -\sin x & -\cos x \end{vmatrix} - \sin x \begin{vmatrix} e^x & -\sin x \\ e^x & -\cos x \end{vmatrix} + \cos x \begin{vmatrix} e^x & \cos x \\ e^x & -\sin x \end{vmatrix}$$

Computing each determinant:

$$\begin{vmatrix} \cos x & -\sin x \\ -\sin x & -\cos x \end{vmatrix} = (-\cos^2 x - \sin^2 x) = -1$$

$$\begin{vmatrix} e^x & -\sin x \\ e^x & -\cos x \end{vmatrix} = (-\cos x e^x + \sin x e^x) = e^x(\sin x - \cos x)$$

$$\begin{vmatrix} e^x & \cos x \\ e^x & -\sin x \end{vmatrix} = (-\sin x e^x - \cos x e^x) = -e^x(\sin x + \cos x)$$

Substituting these values of determinant, we have

$$\begin{aligned}
W &= e^x(-1) - \sin x \cdot e^x(\sin x - \cos x) + \cos x \cdot (-e^x(\sin x + \cos x)) \\
&= -e^x - e^x \sin x(\sin x - \cos x) - e^x \cos x(\sin x + \cos x) \\
&= -e^x(\sin^2 x + \cos^2 x + 1)
\end{aligned}$$

$$W = -2e^x \neq 0$$

Since $W(x) \neq 0$, the functions are linearly independent.

(e) $\{e^x, e^{5+x}\}$

$$f_1 = e^x, \quad f_2 = e^{5+x} = e^5 e^x$$

$$f'_1 = e^x, \quad f'_2 = e^5 e^x$$

$$\begin{aligned}
W &= \begin{vmatrix} e^x & e^5 e^x \\ e^x & e^5 e^x \end{vmatrix} \\
&= e^x \cdot e^5 e^x - e^5 e^x \cdot e^x \\
&= 0
\end{aligned}$$

From a theorem, we know that:

For a homogeneous linear differential equation

$$y'' + p(x)y' + q(x)y = 0.$$

where $p(x)$ and $q(x)$ are continuous on an interval I . Then, any two solutions y_1 and y_2 of the differential equations are linearly dependent iff Wronskian $W(y_1, y_2) = 0$ at some points $x_0 \in I$.

Now, consider the homogeneous differential equation with continuous coefficients.

$$y'' + e^x y' - (e^x + 1)y = 0.$$

We can easily verify that e^x and e^{5+x} are the solution of the above differential equation and we have $W(e^x, e^{5+x}) = 0$.

Therefore, the set $\{e^x, e^{5+x}\}$ is linearly dependent.

(f) $\{x, x^2, x^3\}$

$$f_1 = x, \quad f_2 = x^2, \quad f_3 = x^3$$

$$f'_1 = 1, \quad f'_2 = 2x, \quad f'_3 = 3x^2$$

$$f''_1 = 0, \quad f''_2 = 2, \quad f''_3 = 6x$$

$$W = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

$$\implies W = x \begin{vmatrix} 2x & 3x^2 \\ 2 & 6x \end{vmatrix} - x^2 \begin{vmatrix} 1 & 3x^2 \\ 0 & 6x \end{vmatrix} + x^3 \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix}$$

Computing each determinant:

$$\begin{vmatrix} 2x & 3x^2 \\ 2 & 6x \end{vmatrix} = 12x^2 - 6x^2 = 6x^2$$

$$\begin{vmatrix} 1 & 3x^2 \\ 0 & 6x \end{vmatrix} = 6x$$

$$\begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = 2$$

Substituting these values:

$$W = x(6x^2) - x^2(6x) + x^3(2)$$

$$= 6x^3 - 6x^3 + 2x^3 = 2x^3$$

Since $W(x) \neq 0$, the functions are linearly independent.

(g) $\{\ln x, \ln x^2, x\}$

$$f_1 = \ln x, \quad f_2 = \ln x^2, \quad f_3 = x$$

Since $\ln x^2 = 2 \ln x$, we can rewrite f_2 as:

$$f_2 = 2 \ln x$$

$$f'_1 = \frac{1}{x}, \quad f'_2 = 2 \frac{1}{x} = \frac{2}{x}, \quad f'_3 = 1$$

$$f''_1 = -\frac{1}{x^2}, \quad f''_2 = -\frac{2}{x^2}, \quad f''_3 = 0$$

$$W = \begin{vmatrix} \ln x & 2 \ln x & x \\ \frac{1}{x} & \frac{2}{x} & 1 \\ -\frac{1}{x^2} & -\frac{2}{x^2} & 0 \end{vmatrix} = 0$$

Now, to show whether or not the given set of functions is linearly independent, Consider the first two functions, i.e., $\ln x, \ln x^2$.

The Wronskian

$$W(\ln x, \ln x^2) = \begin{vmatrix} \ln x & 2 \ln x \\ \frac{1}{x} & \frac{2}{x} \end{vmatrix} = 0$$

From a theorem, we know that:

For a homogeneous linear differential equation

$$y'' + p(x)y' + q(x)y = 0.$$

where $p(x)$ and $q(x)$ are continuous on an interval I . Then, any two solutions y_1 and y_2 of the differential equations are linearly dependent iff Wronskian $W(y_1, y_2)=0$ at some points $x_0 \in I$.

Now, consider the homogeneous differential equation with continuous coefficients over an interval I not containing 0 and e.

$$y'' - \frac{1}{x(\ln x - 1)} y' + \frac{1}{x^2(\ln x - 1)} y = 0.$$

We can easily verify that $\ln x$, and $\ln x^2$ are the solutions of the above differential equation, and we also have the Wronskian $W(\ln x, \ln x^2)=0$

Therefore, the two functions $\ln x$, and $\ln x^2$ are linearly dependent. Hence the set $\{\ln x, \ln x^2, x\}$ is linearly dependent.

(h) $\{x, x^2, \sin x\}$

$$f_1 = x, \quad f_2 = x^2, \quad f_3 = \sin x$$

$$f'_1 = 1, \quad f'_2 = 2x, \quad f'_3 = \cos x$$

$$f''_1 = 0, \quad f''_2 = 2, \quad f''_3 = -\sin x$$

$$W = \begin{vmatrix} x & x^2 & \sin x \\ 1 & 2x & \cos x \\ 0 & 2 & -\sin x \end{vmatrix}$$

$$\Rightarrow W = x \begin{vmatrix} 2x & \cos x \\ 2 & -\sin x \end{vmatrix} - x^2 \begin{vmatrix} 1 & \cos x \\ 0 & -\sin x \end{vmatrix} + \sin x \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix}$$

Now computing each determinant

$$\begin{vmatrix} 2x & \cos x \\ 2 & -\sin x \end{vmatrix} = (2x)(-\sin x) - (\cos x)(2) = -2x \sin x - 2 \cos x.$$

$$\begin{vmatrix} 1 & \cos x \\ 0 & -\sin x \end{vmatrix} = (1)(-\sin x) - (\cos x)(0) = -\sin x.$$

$$\begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = (1)(2) - (2x)(0) = 2.$$

Substituting these back into the expansion:

$$\begin{aligned} W(x) &= x(-2x \sin x - 2 \cos x) - x^2(-\sin x) + \sin x \cdot 2 \\ &= -2x^2 \sin x - 2x \cos x + x^2 \sin x + 2 \sin x \\ &= -x^2 \sin x - 2x \cos x + 2 \sin x. \end{aligned}$$

The Wronskian is:

$$W(x) = -x^2 \sin x - 2x \cos x + 2 \sin x,$$

and since it is not identically zero, the functions $\{x, x^2, \sin x\}$ are linearly independent.

(i) $\{e^{mx}, xe^{mx}, x^2e^{mx}\}$

For $f_1(x) = e^{mx}$:

$$f_1'(x) = m e^{mx}, \quad f_1''(x) = m^2 e^{mx}.$$

For $f_2(x) = x e^{mx}$:

$$f_2'(x) = e^{mx} + mx e^{mx} = e^{mx}(1 + mx),$$

$$f_2''(x) = \frac{d}{dx} [e^{mx}(1 + mx)] = m e^{mx}(1 + mx) + m e^{mx} = e^{mx}(2m + m^2 x).$$

For $f_3(x) = x^2 e^{mx}$:

$$f_3'(x) = 2x e^{mx} + x^2 m e^{mx} = e^{mx}(2x + mx^2),$$

$$f_3''(x) = \frac{d}{dx} [e^{mx}(2x + mx^2)] = m e^{mx}(2x + mx^2) + e^{mx}(2 + 2mx) = e^{mx}(m^2 x^2 + 4mx + 2).$$

Now, the Wronskian is

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ f_1'(x) & f_2'(x) & f_3'(x) \\ f_1''(x) & f_2''(x) & f_3''(x) \end{vmatrix} = \begin{vmatrix} e^{mx} & x e^{mx} & x^2 e^{mx} \\ m e^{mx} & e^{mx}(1 + mx) & e^{mx}(2x + mx^2) \\ m^2 e^{mx} & e^{mx}(2m + m^2 x) & e^{mx}(m^2 x^2 + 4mx + 2) \end{vmatrix}.$$

Notice that each entry has a factor of e^{mx} ; factoring e^{mx} from each column gives

$$W(x) = e^{3mx} \begin{vmatrix} 1 & x & x^2 \\ m & 1 + mx & 2x + mx^2 \\ m^2 & 2m + m^2 x & m^2 x^2 + 4mx + 2 \end{vmatrix}.$$

Expanding this determinant gives a nonzero result, so the functions are linearly independent.

Question 3

We have to show the set of functions $e^{m_1x}, e^{m_2x}, \dots, e^{m_nx}$ is linearly independent, where $m_1, m_2, \dots, m_n$ are distinct real numbers.

Let us define the functions:

$$y_i(x) = e^{m_i x}, \quad \text{for } i = 1, 2, \dots, n$$

The Wronskian $W(y_1, y_2, \dots, y_n)(x)$ is given by the determinant:

$$W(x) = \begin{vmatrix} e^{m_1x} & e^{m_2x} & \dots & e^{m_nx} \\ m_1 e^{m_1x} & m_2 e^{m_2x} & \dots & m_n e^{m_nx} \\ \vdots & \vdots & \ddots & \vdots \\ m_1^{n-1} e^{m_1x} & m_2^{n-1} e^{m_2x} & \dots & m_n^{n-1} e^{m_nx} \end{vmatrix}$$

$$\Rightarrow W(x) = e^{(m_1+m_2+\dots+m_n)x} \cdot \begin{vmatrix} 1 & 1 & \dots & 1 \\ m_1 & m_2 & \dots & m_n \\ \vdots & \vdots & \ddots & \vdots \\ m_1^{n-1} & m_2^{n-1} & \dots & m_n^{n-1} \end{vmatrix}$$

This is a Vandermonde determinant, whose value is:

$$\prod_{1 \leq i < j \leq n} (m_j - m_i)$$

Since the m_i 's are distinct, each term $(m_j - m_i) \neq 0$, so the entire product is non-zero. Hence the Wronskian $W(x) \neq 0$

Therefore the set of functions $e^{m_1x}, e^{m_2x}, \dots, e^{m_nx}$ are linearly independent.

Question 4

Are the following statements true or false?

- (a) If $y_1(x)$ and $y_2(x)$ are linearly independent on an interval I , then they are linearly independent on any interval containing I .
- (b) If $y_1(x)$ and $y_2(x)$ are linearly dependent on an interval I , then they are linearly dependent on any subinterval of I .

Solution:

(a) **Answer: True.**

Suppose $y_1(x)$ and $y_2(x)$ are linearly independent on an interval I . By definition, this means that for any constants c_1 and c_2 , the equation

$$c_1 y_1(x) + c_2 y_2(x) = 0 \quad \text{for all } x \in I$$

implies that $c_1 = 0$ and $c_2 = 0$. Now, let J be any interval containing I (i.e., $J \supset I$). To prove that $y_1(x)$ and $y_2(x)$ are also linearly independent on J , assume for the sake of contradiction that they are not linearly independent on J . Then there exist constants c_1 and c_2 , not both zero, such that

$$c_1 y_1(x) + c_2 y_2(x) = 0 \quad \text{for all } x \in J.$$

Since $I \subset J$, this equation must also hold for all $x \in I$. However on I , the only solution to the equation is $c_1 = 0$ and $c_2 = 0$.

Thus, the assumption that $y_1(x)$ and $y_2(x)$ are linearly dependent on J is false. Therefore, $y_1(x)$ and $y_2(x)$ must be linearly independent on any interval J containing I .

(b) **Answer: True.**

Suppose $y_1(x)$ and $y_2(x)$ are linearly dependent on an interval I . By definition, this means there exist constants c_1 and c_2 , not both zero, such that

$$c_1 y_1(x) + c_2 y_2(x) = 0 \quad \text{for all } x \in I.$$

Now, let J be any subinterval of I , i.e., $J \subset I$. Since J is contained within I , every x in J is also in I . Thus, the same linear combination

$$c_1 y_1(x) + c_2 y_2(x) = 0$$

holds for all $x \in J$, with the same constants c_1 and c_2 , where not both are zero.

This shows that $y_1(x)$ and $y_2(x)$ are linearly dependent on the subinterval J as well.

Question 5

Let $y_1(x)$ and $y_2(x)$ be two solutions of the differential equation

$$y'' + P(x)y' + Q(x)y = 0,$$

where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Are the following statements true or false?

- (a) If $y_1(x)$ and $y_2(x)$ are linearly independent on an interval I , they are linearly independent on any interval J contained in I .
- (b) If $y_1(x)$ and $y_2(x)$ are linearly dependent on an interval I , they are linearly dependent on any interval J contained in or containing I .

Solution:

- (a) **Answer: True.**

If $y_1(x)$ and $y_2(x)$ are linearly independent on I , then their Wronskian is **not identically zero** on I , i.e.,

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0 \quad \text{for some } x \in I. \quad (1)$$

The fundamental theorem of Wronskians states that if the Wronskian of two solutions is identically zero on an interval, then the solutions are **linearly dependent** on the entire interval.

If $W(y_1, y_2)(x)$ is nonzero at least for one point in I , then the solutions remain linearly independent **throughout** I .

Since any subinterval $J \subset I$ inherits the property that $W(y_1, y_2)(x)$ is **not identically zero**, the solutions remain linearly independent on J .

Thus, the given statement is **true**.

(b) **Case 1: Linear Dependence on a Subinterval**

Answer: True.

Suppose $y_1(x)$ and $y_2(x)$ are linearly dependent on the interval I . This means that their Wronskian $W(y_1, y_2)(x)$ is zero for all $x \in I$.

Now, consider any subinterval $J \subset I$. Since $P(x)$ and $Q(x)$ are continuous on I , they are also continuous on J . The Wronskian $W(y_1, y_2)(x)$, which is determined by $y_1(x)$, $y_2(x)$, and their derivatives, will still be zero on any subinterval $J \subset I$.

Therefore, if $y_1(x)$ and $y_2(x)$ are linearly dependent on I , they will also be linearly dependent on any subinterval $J \subset I$.

Case 2: Linear Dependence on a Larger Interval

- If $P(x)$ and $Q(x)$ are continuous on J and $y_1(x)$ and $y_2(x)$ are linearly dependent on I , then their Wronskian will remain zero on J , and they will still be linearly dependent on J .
- If $P(x)$ and $Q(x)$ are not continuous on J , then we cannot immediately conclude that the functions remain linearly dependent on J without further analysis of the behavior of the differential equation on the larger interval.

Thus, when $J \supseteq I$, the statement is true only if $P(x)$ and $Q(x)$ are continuous on J . Otherwise, linear dependence may not necessarily hold on J .

Question 6

Let $y_1(x)$ and $y_2(x)$ be two solutions of the differential equation

$$y'' + P(x)y' + Q(x)y = 0,$$

where $x \in I = (a, b)$, and $P(x)$ and $Q(x)$ are continuous functions on I . Let $W(x)$ be the Wronskian of these two solutions. We aim to show that

$$W'(x) = -P(x)W(x).$$

Additionally, if $W(x_0) = 0$ for some $x_0 \in I$, we will prove that $W(x) = 0$ for each $x \in I$.

Solution :

Let $y_1(x)$ and $y_2(x)$ be two solutions of

$$y'' + P(x)y' + Q(x)y = 0, \tag{1}$$

and let

$$W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x),$$

be their Wronskian. We can compute the derivative $W'(x)$ as

$$\begin{aligned} W'(x) &= y_1'(x)y_2'(x) + y_1(x)y_2''(x) - y_1''(x)y_2(x) - y_1'(x)y_2'(x) \\ &= y_1(x)y_2''(x) - y_1''(x)y_2(x). \end{aligned} \quad (2)$$

Since y_1 and y_2 are solutions of (1), we have

$$y_1''(x) = -P(x)y_1'(x) - Q(x)y_1(x), \quad \text{and} \quad y_2''(x) = -P(x)y_2'(x) - Q(x)y_2(x),$$

and substituting these into (2), we get

$$\begin{aligned} W'(x) &= y_1(x)[-P(x)y_2'(x) - Q(x)y_2(x)] - [-P(x)y_1'(x) - Q(x)y_1(x)]y_2(x) \\ &= -P(x)y_1(x)y_2'(x) - Q(x)y_1(x)y_2(x) + P(x)y_1'(x)y_2(x) + Q(x)y_1(x)y_2(x) \\ &= -P(x)y_1(x)y_2'(x) + P(x)y_1'(x)y_2(x) \\ &= -P(x)[y_1(x)y_2'(x) - y_1'(x)y_2(x)] \\ &= -P(x)W(x). \end{aligned}$$

In other words, the Wronskian satisfies the first order linear equation

$$W'(x) + P(x)W(x) = 0. \quad (3)$$

Next, for the solution of (3) we multiply both sides by the integrating factor $e^{\int P(x) dx}$:

$$e^{\int P(x) dx} W'(x) + e^{\int P(x) dx} P(x)W(x) = 0$$

This simplifies to:

$$\frac{d}{dx} \left(e^{\int P(x) dx} W(x) \right) = 0$$

Integrating both sides of the equation gives:

$$e^{\int P(x) dx} W(x) = C$$

Finally, solving for $W(x)$, we obtain the general solution:

$$W(x) = Ce^{-\int P(x) dx}$$

If $W(x_0) = 0$, then $C = 0$, and hence:

$$W(x) = 0 \quad \text{for all } x \in I.$$

Therefore, if $W(x_0) = 0$ for some $x_0 \in I$, then $W(x) = 0$ for all $x \in I$.

Question 7

Find the general solution of the following second-order homogeneous differential equations:

(a) $y'' + 3y' - 10y = 0$.

Solution: Assume a solution of the form:

$$y = e^{mx}.$$

Substituting into the differential equation:

$$m^2 e^{mx} + 3m e^{mx} - 10e^{mx} = 0.$$

Since $e^{mx} \neq 0$, we obtain the characteristic equation:

$$m^2 + 3m - 10 = 0.$$

$$(m + 5)(m - 2) = 0.$$

Solving for m :

$$m_1 = -5, \quad m_2 = 2.$$

Thus, the general solution is:

$$y(x) = Ae^{-5x} + Be^{2x}.$$

where A and B are arbitrary constants.

(b). $2y'' - 3y' = 0$.

Solution: Assume a solution of the form:

$$y = e^{mx}.$$

Substituting into the differential equation we get:

$$2m^2 e^{mx} - 3m e^{mx} = 0.$$

Since $e^{mx} \neq 0$, we obtain the characteristic equation:

$$2m^2 - 3m = 0.$$

$$m(2m - 3) = 0.$$

Therefore, we get:

$$m_1 = 0, \quad m_2 = \frac{3}{2}.$$

Thus, the general solution is:

$$y(x) = C_1 + C_2 e^{\frac{3}{2}x}.$$

where C_1 and C_2 are arbitrary constants.

Question 8

Find the curve $y(x)$ through the origin for which $y'' = y'$ and the tangent at the origin is $y = x$.

Solution: We are given the second-order differential equation:

$$y'' = y'$$

For solving it rewriting the equation:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{dy}{dx} \\ \implies \frac{d^2y}{dx^2} - \frac{dy}{dx} &= 0\end{aligned}$$

This represents a linear homogeneous differential equation with the characteristic equation:

$$\begin{aligned}m^2 - m &= 0 \\ \implies m(m - 1) &= 0\end{aligned}$$

The roots are $m = 0$ and $m = 1$,
So the general solution is:

$$y(x) = C_1e^x + C_2 \tag{2}$$

Applying the initial conditions,

Condition 1: The curve passes through the origin

$$y(0) = 0$$

Substituting $x = 0$:

$$C_1e^0 + C_2 = 0 \Rightarrow C_1 + C_2 = 0$$

Thus, $C_2 = -C_1$, which on substituting in Eq.(2) we get:

$$y(x) = C_1(e^x - 1) \tag{3}$$

Condition 2: The tangent at the origin is $y = x$.

Since the tangent line at the origin is $y = x \implies y'(x) = 1$, we get:

$$y'(0) = 1 \tag{4}$$

The derivative of Eq.(3) is:

$$y'(x) = C_1e^x \tag{5}$$

Substituting $x = 0$ we get,

$$y'(0) = C_1e^0 \tag{6}$$

Therefore from Eq.(4) and (6) we get

$$C_1e^0 = 1$$

$$\implies C_1 = 1$$

Thus, the solution Eq.(3) becomes:

$$y(x) = e^x - 1$$

Thus, the required curve is:

$$y(x) = e^x - 1$$

This function passes through the origin and has the correct tangent line $y = x$ at $x = 0$.

Solution of 9:

Given the differential equation of the form

$$y'' + ay' + by = 0 \tag{7}$$

where a and b are constants.

And given that (a) e^{2x} , 1 (b) $e^{(\alpha+i\beta)x}$, $e^{(\alpha-i\beta)x}$ be the solution of Eq. (7). We have to find values of a & b and the differential equation corresponding to solutions.

(a) Differential equation with solutions e^{2x} and 1

Step 1: Plug in $y = 1$.

$$y = 1 \implies y' = 0, \quad y'' = 0.$$

Substitute into the differential equation:

$$0 + a \cdot 0 + b \cdot 1 = b = 0.$$

Hence, we find

$$b = 0.$$

Step 2: Plug in $y = e^{2x}$.

$$y = e^{2x} \implies y' = 2e^{2x}, \quad y'' = 4e^{2x}.$$

Substitute y, y', y'' into the differential equation (using $b = 0$):

$$4e^{2x} + a \cdot (2e^{2x}) + 0 \cdot e^{2x} = e^{2x}(4 + 2a) = 0.$$

Since $e^{2x} \neq 0$ for all x , we must have

$$4 + 2a = 0 \implies 2a = -4 \implies a = -2.$$

Now, putting the values of a and b in Eq. (7), we get:

$$\boxed{y'' - 2y' = 0.}$$

(b) Differential equation with solutions $e^{(\alpha+i\beta)x}$ and $e^{(\alpha-i\beta)x}$

We know that if e^{rx} is a solution of Eq. (7), then r must be a root of the characteristic equation

$$r^2 + ar + b = 0, \quad (8)$$

which is a quadratic equation.

We are given two solutions whose exponents (roots) are:

$$r_1 = \alpha + i\beta, \quad r_2 = \alpha - i\beta.$$

For a quadratic equation (8), we use the relationships:

$$r_1 + r_2 = -a, \quad r_1 r_2 = b.$$

Sum of roots.

$$r_1 + r_2 = \alpha + i\beta + \alpha - i\beta = 2\alpha.$$

Hence

$$-a = 2\alpha \implies a = -2\alpha.$$

Product of roots.

$$r_1 r_2 = (\alpha + i\beta) \cdot (\alpha - i\beta) = \alpha^2 + \beta^2.$$

Thus

$$b = \alpha^2 + \beta^2.$$

Putting it all together, the desired differential equation is:

$$\boxed{y'' - 2\alpha y' + (\alpha^2 + \beta^2) y = 0.}$$

Solution of 10:

We have to find a homogeneous linear differential equation (on $(0, \infty)$) whose general solution is

$$y(t) = c_1 t^2 e^t + c_2 t^3 e^t.$$

Since, we have two linearly independent solutions

$$y_1(t) = t^2 e^t \quad \text{and} \quad y_2(t) = t^3 e^t.$$

We know a second-order linear differential equation of the form

$$y'' + a_1(t) y' + a_0(t) y = 0, \quad (9)$$

valid on $(0, \infty)$, for which both y_1 and y_2 are solutions.

Step 1: Compute derivatives of $y_1(t) = t^2 e^t$

$$\begin{aligned} y_1(t) &= t^2 e^t, \\ y_1'(t) &= \frac{d}{dt}(t^2 e^t) = e^t (2t + t^2), \\ y_1''(t) &= \frac{d}{dt}(e^t (2t + t^2)) = e^t (2t + t^2) + e^t (2 + 2t) \\ &= e^t (2 + 4t + t^2). \end{aligned}$$

Thus,

$$y_1''(t) = e^t [2 + 4t + t^2].$$

Step 2: Compute derivatives of $y_2(t) = t^3 e^t$

$$\begin{aligned} y_2(t) &= t^3 e^t, \\ y_2'(t) &= \frac{d}{dt}(t^3 e^t) = e^t (3t^2 + t^3), \\ y_2''(t) &= \frac{d}{dt}(e^t (3t^2 + t^3)) = e^t (3t^2 + t^3) + e^t (6t + 3t^2) \\ &= e^t (6t + 6t^2 + t^3). \end{aligned}$$

Thus,

$$y_2''(t) = e^t [6t + 6t^2 + t^3].$$

For $y_1(t) = t^2 e^t$:

$$\begin{aligned} y_1'' + a_1(t)y_1' + a_0(t)y_1 &= e^t [2 + 4t + t^2] + a_1(t) e^t [2t + t^2] + a_0(t) e^t t^2 \\ &= e^t ([2 + 4t + t^2] + a_1(t)[2t + t^2] + a_0(t)t^2) = 0. \end{aligned}$$

Since $e^t \neq 0$, we have

$$[2 + 4t + t^2] + a_1(t)[2t + t^2] + a_0(t)t^2 = 0. \quad (10)$$

For $y_2(t) = t^3 e^t$:

$$\begin{aligned} y_2'' + a_1(t)y_2' + a_0(t)y_2 &= e^t [6t + 6t^2 + t^3] + a_1(t) e^t [3t^2 + t^3] + a_0(t) e^t t^3 \\ &= e^t ([6t + 6t^2 + t^3] + a_1(t)[3t^2 + t^3] + a_0(t)t^3) = 0. \end{aligned}$$

Thus,

$$[6t + 6t^2 + t^3] + a_1(t)[3t^2 + t^3] + a_0(t)t^3 = 0. \quad (11)$$

Now, solving Eqs. (10) and (11) which are linear systems in $a_1(t)$ and $a_0(t)$.

$$a_1(t) = -\frac{4}{t} - 2 \quad \text{and} \quad a_0(t) = 1 + \frac{4}{t} + \frac{6}{t^2}.$$

After putting these values in Eqn (9)-

$$y'' - \left(\frac{4}{t} + 2\right)y' + \left(1 + \frac{4}{t} + \frac{6}{t^2}\right)y = 0.$$

or

$$t^2 y'' - (4t + 2t^2)y' + (t^2 + 4t + 6)y = 0.$$

And, there is no constant-coefficient homogeneous ODE whose general solution is

$$c_1 t^2 e^t + c_2 t^3 e^t.$$

Solution of 11

We have to solve the given differential equation of the form

$$4y'''' - 12y''' - y'' + 27y' - 18y = 0. \quad (12)$$

Characteristic equation of Eq. (12) is

$$4r^4 - 12r^3 - r^2 + 27r - 18 = 0 \quad (13)$$

Let us check that $r = 1$ is a solution of the Eq. (13) and substituting in the Eq. (13), we get

$$4 - 12 - 1 + 27 - 18 = 0.$$

Thus, $r = 1$ is the solution of the Eq. (13).

From Eq. (13)

$$(r - 1)(4r^3 - 8r^2 - 9r + 18) = 0$$

$$(r - 1)(4r^2 - 9)(r - 2) = 0$$

The r values are $-3/2, 1, 3/2$, and 2 .

Now, the solution of Eq. (12) is

$$y(x) = Ae^x + Be^{2x} + Ce^{-3/2x} + De^{3/2x}.$$

12.Solution

Substituting $y = e^{mx}$, we obtain the characteristic equation as

$$m^3 + 3m^2 - 4 = 0$$

or

$$(m - 1)(m^2 + 4m + 4) = 0$$

or

$$(m-1)(m+2)^2 = 0$$

The roots of this equation are $m = 1, -2, -2$.
The general solution is,

$$y(x) = Ae^x + (Bx + C)e^{-2x}$$

Substituting into the initial conditions, we get

$$y(0) = 1 = A + C$$

$$y'(x) = Ae^x + Be^{-2x} - 2(Bx + C)e^{-2x}$$

$$y'(0) = 0 = A + B - 2C$$

$$y''(x) = Ae^x - 4Be^{-2x} + 4(Bx + C)e^{-2x}$$

$$y''(0) = \frac{1}{2} = A - 4B + 4C$$

The solution of the system is $A = \frac{1}{2}$, $B = \frac{1}{2}$, and $C = \frac{1}{2}$. The particular solution is:

$$y(x) = \frac{e^x + (x+1)e^{-2x}}{2}$$

13.Solution

Given the differential equation:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = r(x) \quad (1)$$

$$y_1 = 1 + e^{x^2}$$

$$y_2 = 1 + xe^{x^2}$$

$$y_3 = (1+x)e^{x^2} - 1$$

are solutions of equation (1).

Since $y_2(x)$ and $y_1(x)$ are solutions of equation (1),

$$y_2 - y_1$$

will be a solution of the corresponding homogeneous equation:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (2)$$

Thus,

$$(x - 1)e^{x^2}$$

is a solution of (2).

Similarly,

$$y_3 - y_2 = e^{x^2} - 2$$

is a solution of (2).

Since, $(x - 1)e^{x^2}$ and $e^{x^2} - 2$ are linearly independent (L.I.).

The general solution of (2) can be written as:

$$C_1(x - 1)e^{x^2} + C_2(e^{x^2} - 2)$$

where $y_1 = 1 + e^{x^2}$ is a particular solution of (1).

Thus, the general solution of (1) is,

$$C_1(x - 1)e^{x^2} + C_2(e^{x^2} - 2) + (1 + e^{x^2})$$

14.Solution

Symbolic form of the equation: $(D^2 - 4)y = \cosh(2x - 1) + 3^x$.

Its auxiliary equation becomes,

$$(D^2 - 4) = 0 \implies D^2 = 4 \quad \text{or} \quad D = \pm 2$$

So that

$$y_{C.F.}(x) = c_1 e^{2x} + c_2 e^{-2x}$$

And

$$y(x)_{P.I} = \frac{1}{D^2 - 4} \cosh(2x - 1) + \frac{1}{D^2 - 4} 3^x$$

implying

$$\begin{aligned} y_{P.I} &= \frac{1}{D^2 - 4} \left[\frac{e^{(2x-1)} + e^{-(2x-1)}}{2} \right] + \frac{1}{D^2 - 4} 3^x \\ &= \frac{e^{-1}}{2} \frac{1}{D^2 - 4} e^{2x} + \frac{e}{2} \frac{1}{D^2 - 4} e^{-2x} + \frac{e^{\log 3^x}}{D^2 - 4} \quad [3^x \text{ is comparable to } a^x = e^{x \log a}] \\ &= \frac{1}{2e} x \frac{1}{2D} e^{2x} + \frac{e}{2} x \frac{1}{2D} e^{-2x} + \frac{e^{(\log 3)x}}{(\log 3)^2 - 4} \quad [3^x = e^{x \cdot \log 3} = e^{(\log 3)x}] \\ &= \frac{x e^{2x}}{2e \cdot 4} + \frac{x e \cdot e^{-2x}}{2 \cdot 4} + \frac{3^x}{(\log 3)^2 - 4} \quad (\text{Rewrite, } e^{(\log 3)x} = 3^x) \end{aligned}$$

$$\begin{aligned}
&= \frac{x}{4} \left[\frac{e^{2x-1} - e^{-(2x-1)}}{2} \right] + \frac{3^x}{(\log 3)^2 - 4} \\
&= \frac{x}{4} \sinh(2x - 1) + \frac{3^x}{(\log 3)^2 - 4}
\end{aligned}$$

Therefore, the complete solution is:

$$y = c_1 e^{2x} + c_2 e^{-2x} + \frac{x}{4} \sinh(2x - 1) + \frac{3^x}{(\log 3)^2 - 4}$$

15.Solution

Symbolic form becomes

$$(D^3 - D^2 - 6D)y = 1 + x^2$$

Auxiliary Equation,

$$D^3 - D^2 - 6D = 0$$

Implying

$$D(D + 2)(D - 3) = 0 \quad \text{or} \quad D = 0, 3, -2$$

Hence

$$y_{C.F.} = c_1 + c_2 e^{3x} + c_3 e^{-2x}$$

And

$$\begin{aligned}
y_{P.I.} &= \frac{1}{D^3 - D^2 - 6D} (1 + x^2) = \frac{1}{-6D} \left(1 + \frac{D}{6} - \frac{D^2}{6} \right) (1 + x^2) \\
&= -\frac{1}{6D} \left[1 + \left(\frac{D}{6} - \frac{D^2}{6} \right) \right]^{-1} (1 + x^2) \\
&= -\frac{1}{6D} \left[1 - \left(\frac{D}{6} - \frac{D^2}{6} \right) + \left(\frac{D}{6} - \frac{D^2}{6} \right)^2 + \dots \right] (1 + x^2) \\
&= -\frac{1}{6D} \left[1 - \left(\frac{D}{6} - \frac{D^2}{6} \right) + \frac{D^2}{36} \right] (1 + x^2) \\
&= -\frac{1}{6D} \left[(1 + x^2) - \frac{D}{6} (1 + x^2) + \frac{7}{36} D^2 (1 + x^2) \right] \\
&= -\frac{1}{6D} \left[(1 + x^2) - \frac{x}{3} + \frac{7}{36} (2) \right] \\
&= -\frac{1}{6D} \left[x^2 - \frac{x}{3} + \frac{25}{8} \right] \\
&= -\frac{1}{6} \left[\frac{x^3}{3} - \frac{1}{3} x^2 + \frac{25}{18} x \right]
\end{aligned}$$

$$= -\frac{1}{108}(6x^3 - 3x^2 + 25x)$$

Therefore,

$$y(x) = (c_1 + c_2 e^{3x} + c_3 e^{-2x}) - \frac{1}{108}(6x^3 - 3x^2 + 25x)$$